# ROHAN K. VARIA


*Network & System Engineer·/ Electronics & Communication Engineering*

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## PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering (ECE) student aspiring to build a career as a Network and System Engineer. Strong foundation in networking concepts, system administration, and IT infrastructure. Hands-on experience through internship and lab practice in configuring networks, troubleshooting systems, and working with Windows/Linux environments. Eager to contribute technical skills and continuously learn in a dynamic IT environment.

## CORE SKILLS & TOOLS

| Networking | System Administration | Tools | Services | Virtualization | Scripting Basics | Extra |
|---|---|---|---|---|---|---|
| **TCP/IP, OSI Model, Routing & Switching** | **Windows Server, Linux Basics** | **Cisco Packet Tracer, Wireshark** | **DHCP, DNS, FTP, SSH** | **Hyper-V / VirtualBox** | **Shell / Command Line** | **Troubleshooting & System Support**<br><br>**Basic Cybersecurity Concepts** |

★ **AI Tools Proficiency:** *Python (Scripting) / Bash, ChatGPT / Google Gemini, Claude AI / Perplexity AI, GitHub Copilot, Workflow Automation (n8n), PowerShell / SSH*

## WORK EXPERIENCE

### PLC Engineer Intern – Can Ends India Ltd.

*Duration: 15 Days*

- Gained hands-on exposure to **PLC-based industrial automation systems** in a manufacturing environment.
- Assisted in **basic PLC programming (Ladder Logic)** and understanding the control flow of automated machines.
- Supported engineers in **troubleshooting PLC-controlled equipment** and identifying faults.
- Observed and participated in **preventive maintenance activities** to ensure smooth machine operations.
- Learned about **control panels, sensors, actuators, and industrial electrical components**.
- Collaborated with the technical team to understand **real-time production processes and automation workflows**.

## FEATURED PROJECTS (LEARNING-DRIVEN PORTFOLIO)

**Campus Network Design & Simulation** — *Cisco Packet Tracer · Subnetting · Routing Protocols*

- **Context:** Designed a scalable network architecture for a multi-building campus environment as part of ECE coursework.
- Implemented OSPF routing and NAT to provide efficient data flow and secure internet access.
- **AI-Assisted:** Used **Claude AI** to calculate complex VLSM subnetting schemes and verify firewall access control lists (ACLs).
- **Key Learning:** Mastered end-to-end network planning, from physical layer considerations to logical addressing and security.

**Smart LPG Leakage Detector & Monitoring System** (In College) — *Arduino · MQ-5 Sensor · ESP8266 · IoT Protocols*

- **Dataset / Context:** Designed and prototyped an IoT-based safety system to detect LPG leaks and prevent fire hazards in residential or industrial environments.
- Implemented a real-time monitoring system using an MQ-5 gas sensor and an ESP8266 Wi-Fi module to send instant alerts to a centralized network dashboard.
- **AI-Assisted:** Used **ChatGPT** to write and debug C++ code for sensor calibration and **Claude AI** to design the logic for an automated solenoid valve shut-off system.
- **Key Learning:** Gained hands-on experience in sensor data acquisition, wireless networking protocols (MQTT/HTTP), and designing fail-safe hardware systems.

**Humanized Pick-and-Place Robotic Arm** (In College) — *Servo Motors · Microcontrollers · Kinematics · Python.*

- **Dataset / Context:** Built a 4-degree-of-freedom (4-DOF) robotic arm capable of identifying and moving objects with human-like precision for warehouse automation scenarios.
- Integrated high-torque servo motors and a mechanical gripper, controlled via a custom-coded interface to handle delicate payloads.
- **AI-Assisted:** Leveraged **GitHub Copilot** to optimize the inverse kinematics algorithms and used **Google Gemini** to simulate stress tests on the structural design before physical assembly.
- **Key Learning:** Developed a deep understanding of control systems, mechanical-to-electrical interfacing, and real-time response optimization for automation.

## EDUCATION

| | |
|---|---|
| Degree — B.E. Electronics and Communication, GEC Bharuch / GTU / | *2022-2027 \| Pursuing:* |
| Diploma \| Electronics and Communication, MSU Baroda | *2022 \| Percentage: 66.15 %* |
| $10^{th}$ Std. \| Jay Ambe Vidayalaya, Vadodara | *2018 \| Percentage: 58 %* |